Q.P. Code: 54322

Third Semester B.Com. Degree Examination,
December 2025 / January 2026
(SEP) 2024-25
COMMERCE
BUSINESS STATISTICS

Time : 3 Hrs | Max. Marks : 80

SECTION - A

Answer any TEN sub questions. Each carries TWO marks. **(10x2=20)**

1. a) Define Statistics.
   b) Mention the types of Collection of Data.
   c) What is Tabulation?
   d) Mention the 2 properties of Arithmetic Mean.
   e) Find the Range : 20, 25, 80, 44, 96, 100 & it's coefficient.
   f) Define Standard deviation.
   g) Define Correlation.
   h) What is Regression?
   i) What is Classification?
   j) Give the meaning of Histogram.
   k) State any two differences between correlation and regression.
   l) Write the formula of Probable error.

SECTION - B

Answer any SIX questions. Each carries 5 marks. **(6x5=30)**

2. Briefly explain the types of classification.
3. Briefly explain the different methods of Collecting Primary Data.
4. From the following data calculate quartic deviation and it's coefficient.

X	:	20	60	80	100	120	140	160	200
f	:	25	48	60	57	29	10	8	2

5. Find the Geometric Mean.

   X : 20, 350, 450, 600, 861, 958

6. Draft a blank table to show the distribution of an employees in an office according to sex salary grade (below 3000, 3000 to 5000 & 5000 above) designation (Supervisors, Assistant, Clerk)

7. Obtain two regression equation from the following data & predict the value of X when Y = 7.

$N = 20$, $\bar{X} = 4$, $\bar{Y} = 2$, $\Sigma X^2 = 1680$, $\Sigma Y^2 = 320$, $\Sigma dxdy = 480$

8. From the following data prepare sub-divided bar diagram.

Items	Family A	Family B
Food	1,400	1,600
H.R.	1,200	1,600
Cloth	700	800
Miscellaneous	1,000	1,000
Savings	700	2,400
Total	**5,000**	**8,000**

9. Construct the frequency distribution table from the following data with class intervals 0-10, 10-20 & so on.

18	35	46	22	07	11	03	12	20	32
17	21	13	29	24	39	23	36	27	43
25	49	10	04	14	26	08	16	28	41

SECTION - C

Answer any THREE questions. Each carries 10 marks. **(3x10=30)**

10. Calculate Mean, Median and Mode from the following data.

C-I :	10-19	20-29	30-39	40-49	50-59	60-69	70-79	80-89
f :	10	18	8	6	4	2	5	7

11. Calculate Std. Deviation and Co-efficient of Variation from the following data.

C-I :	10-20	20-30	30-40	40-50	50-60	60-70
f :	10	18	20	6	4	2

12. Calculate the Karl Pearson's Co-efficient of correlation and it's probable error.

X :	8	10	15	17	20	22	24	25
Y :	25	30	32	35	37	40	42	45

13. From the following data obtain two regression equations.

X :	25	28	35	32	31	36	29	38	34	32
Y :	43	46	49	41	36	32	31	30	33	29

14. Explain different methods of collecting primary data with its merits and demerits.

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